

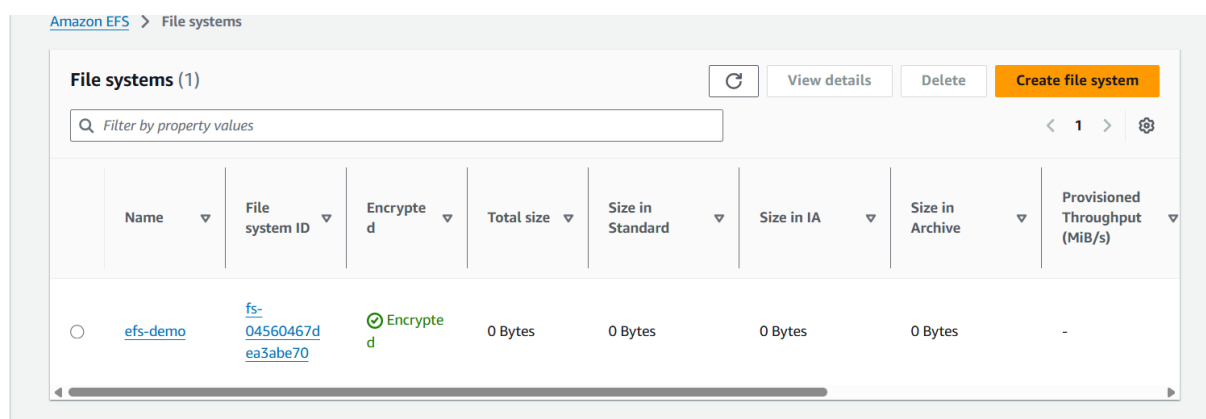
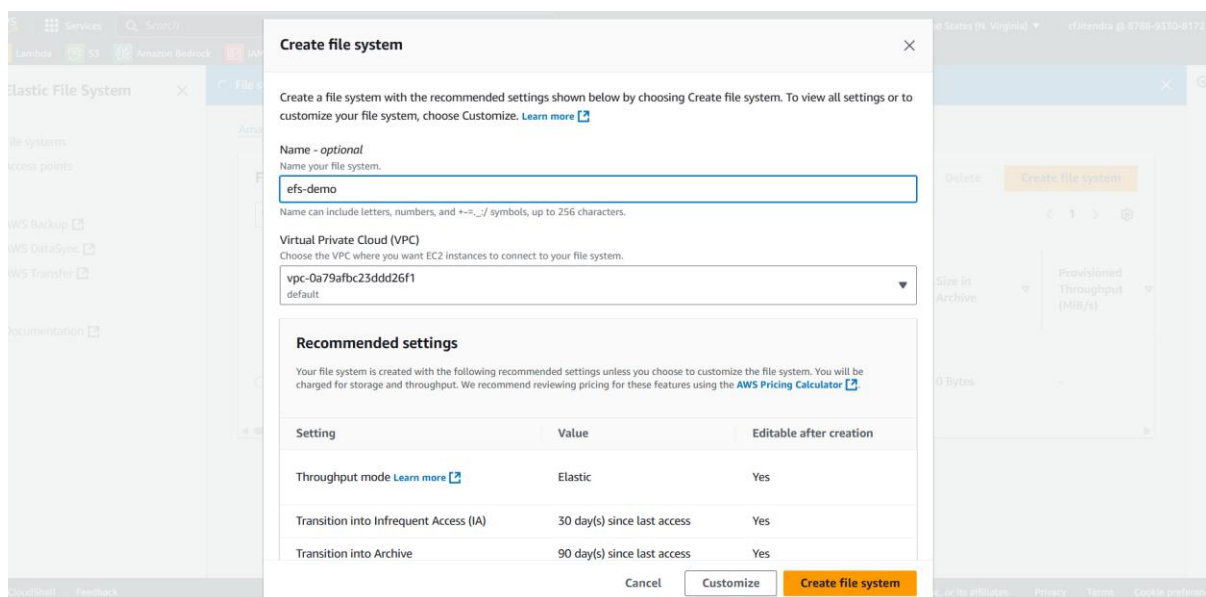
Use Amazon EFS with Amazon EC2 Linux instances

To Begin with the Lab

Summary of the Lab

This lab explains how to create and attach an Amazon EFS to two EC2 Linux instances. First, a file system is created in the EFS console. Then, a Linux EC2 instance is launched with a key pair in .ppk format, and the EFS is attached during configuration. After connecting via PuTTY, a directory is created, and the EFS is mounted using the NFS command. The same process is repeated for a second instance. A test file is created in one instance and verified from the other, confirming shared access.

- Now go to the AWS console
- Navigate to EFS.
- Click on the EFS.
- Go to file system and click create file system.
- Name the file and click create.



- Now we will create a instance and attach the EFS to the instance.

- Open the Amazon EC2 console at [AMAZON EC2](#).
- Choose Launch instance.
- (Optional) Under Name and tags, for Name, enter a name to identify your instance.
- Under Application and OS Images (Amazon Machine Image), choose a Linux operating system, and then for Amazon Machine Image (AMI), select a Linux AMI.

- Under Instance type, for Instance type, select an instance type or keep the default.

- Under Key pair (login), for Key pair name, choose an existing key pair or create a new one.
- Click on the create new key pair.
- Choose the private key file format as '.ppk'.
- Click create key pair.

- Under Network settings, choose Edit (at right), and then for Subnet, select a subnet.

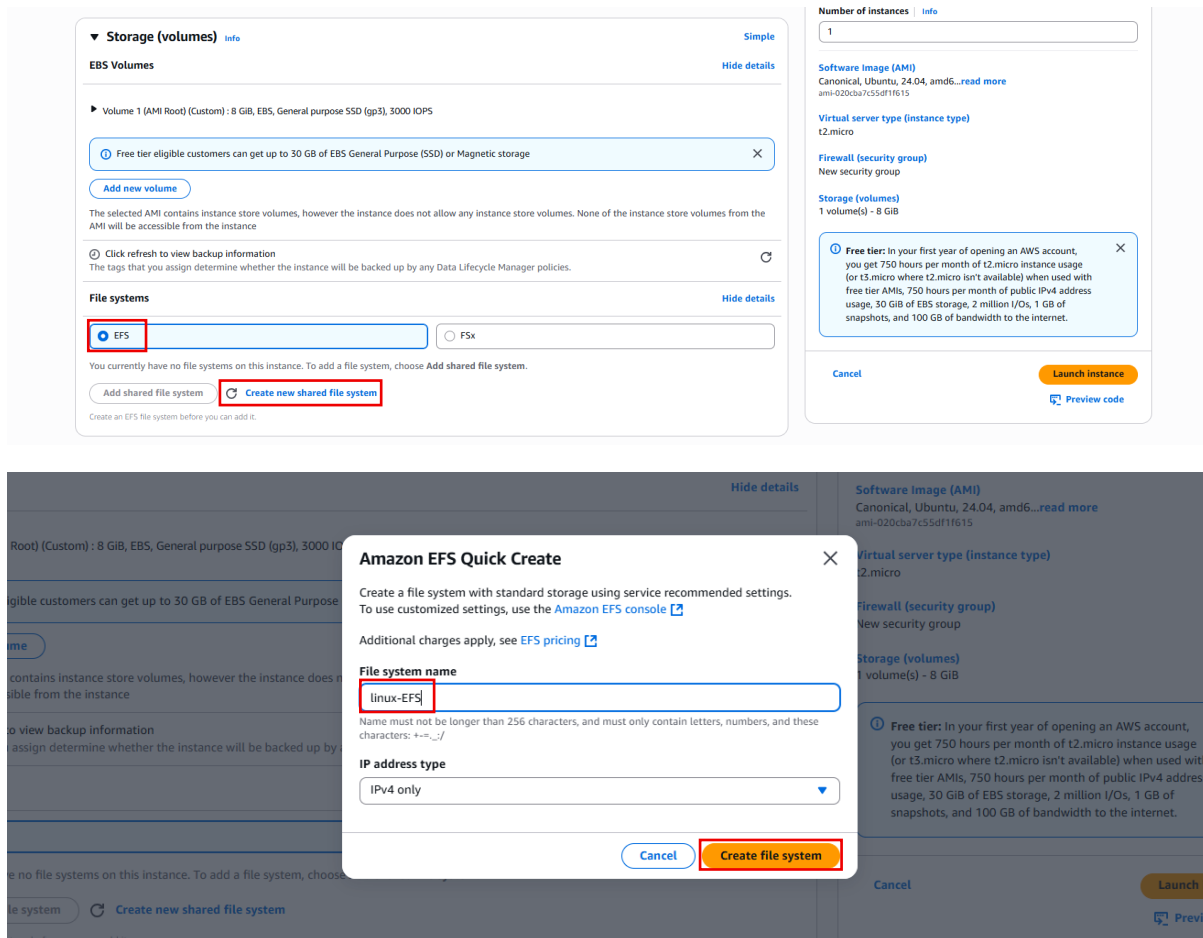
The first screenshot shows the 'Network settings' panel with the 'Edit' button highlighted. The 'Subnet' dropdown is set to 'No preference (Default subnet in any availability zone)'. The 'Summary' panel on the right shows the instance configuration.

The second screenshot shows the 'Network settings' panel with the 'Subnet' dropdown menu open. A red box highlights the 'Create new subnet' link. The 'Summary' panel on the right shows the instance configuration.

- Under Configure storage, choose Edit (at bottom right), and then do the following:
- For File systems, ensure that EFS is selected, and then choose Create new shared file system.
- For File system name enter a name for the Amazon EFS file system, and then choose Create file system.
- Or You can also choose the existing EFS.

The first screenshot shows the 'Configure storage' panel with the 'Edit' button highlighted. The 'Summary' panel on the right shows the instance configuration.

The second screenshot shows the 'Summary' panel with the 'Launch instance' button highlighted. The 'Configure storage' panel on the left shows the instance configuration.



- For Mount point, specify a custom mount point or keep the default.
- To enable access to the file system, select Automatically create and attach security groups. By selecting this checkbox, the following security groups will be automatically created and attached to the instance and the mount targets of the file system:
 - Instance security group – Includes an outbound rule that allows traffic over the NFS 2049port, but includes no inbound rules.
 - File system mount targets security group – Includes an inbound rule that allows traffic over the NFS 2049 port from the instance security group (described above), and an outbound rule that allows traffic over the NFS 2049 port.

Note: - Alternatively, you can manually create and attach the security groups. If you want to manually create and attach the security groups, clear Automatically create and attach the required security groups.

- To automatically mount the shared file system when the instance launches, select Automatically mount shared file system by attaching required user data script. To view the user data that is automatically generated, expand Advanced details, and scroll down to User data.

Note: - If you added user data before selecting this checkbox, the original user data is overwritten by the automatically generated user data.

File systems Hide details

☒ EFS ☐ FSx

▼ Shared file system 1 Remove

File system Info
fs-0a9125489ffe9d7f8
Name: linux-EFS Availability: Regional

Mount point Info
/mnt/efs/fs1

[Add shared file system](#) [Create new shared file system](#)

4 remaining (Up to 5 file systems maximum).

☒ Automatically create and attach security groups
To enable access to the file system, the required security groups will be automatically created and attached to this instance and the selected file system. To manually manage the security groups, clear the checkbox. [Learn more](#).

☒ Automatically mount shared file system by attaching required user data script
Automatically mount your file system by updating your user data to install efs-utils. If you would like to manually mount your file system, clear the checkbox.

Virtual server type (instance type)
t2.micro

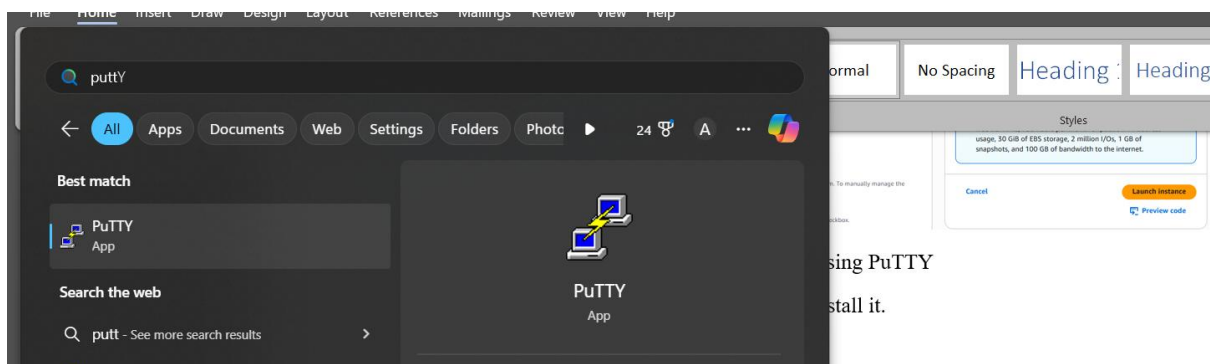
Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

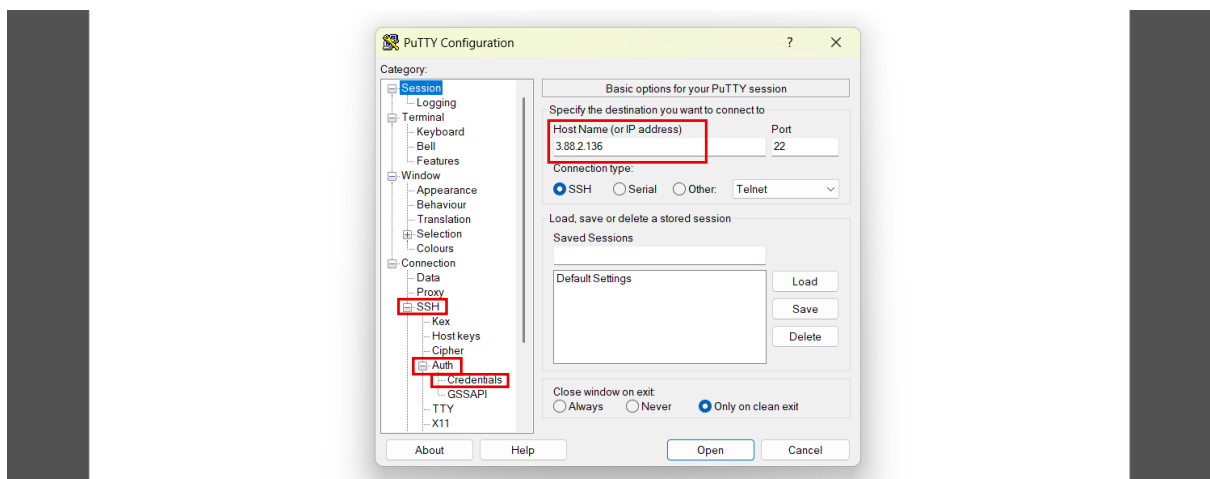
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet.

[Cancel](#) [Launch instance](#) [Preview code](#)

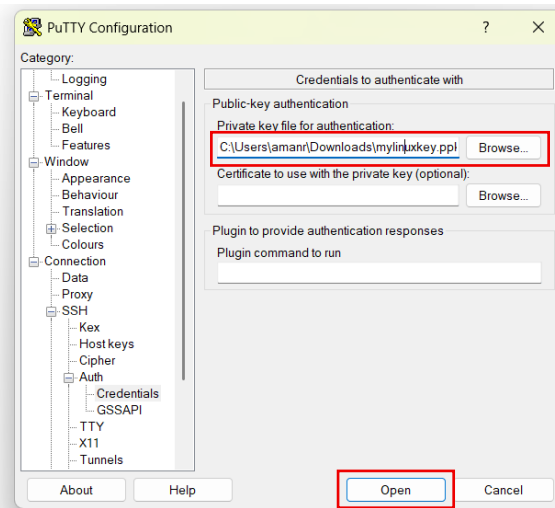
- Now we will connect our Linux virtual machine using PuTTY
- If you haven't downloaded putty download and install it.
- Open PuTTY.



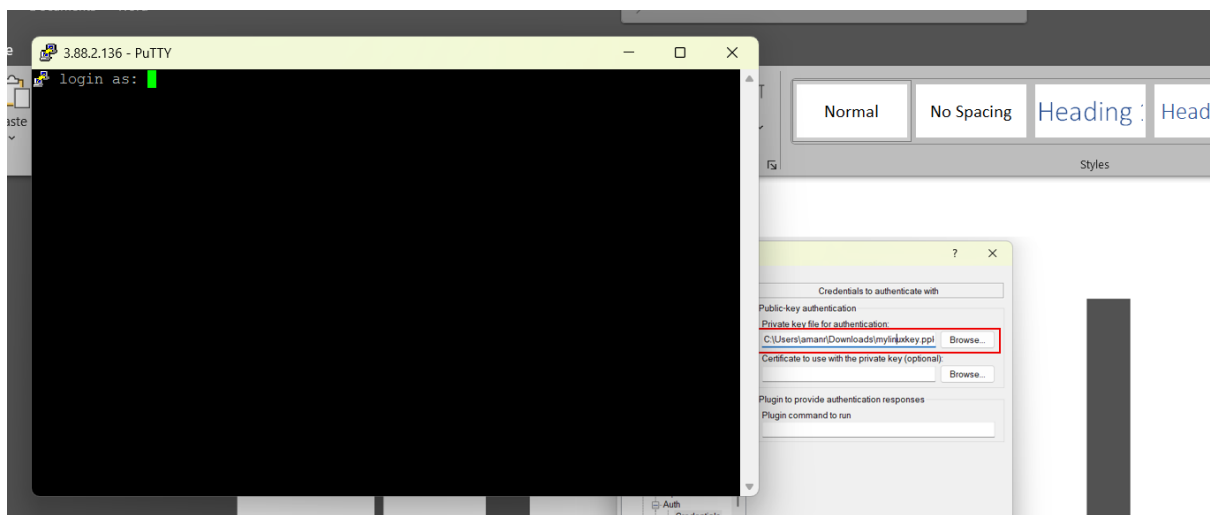
- Enter the public IP address.
- Select the instance and copy the IP address and paste on the Host Name.
- Click on SSH.
- Click on Auth (Authentication).
- Click on Credentials.



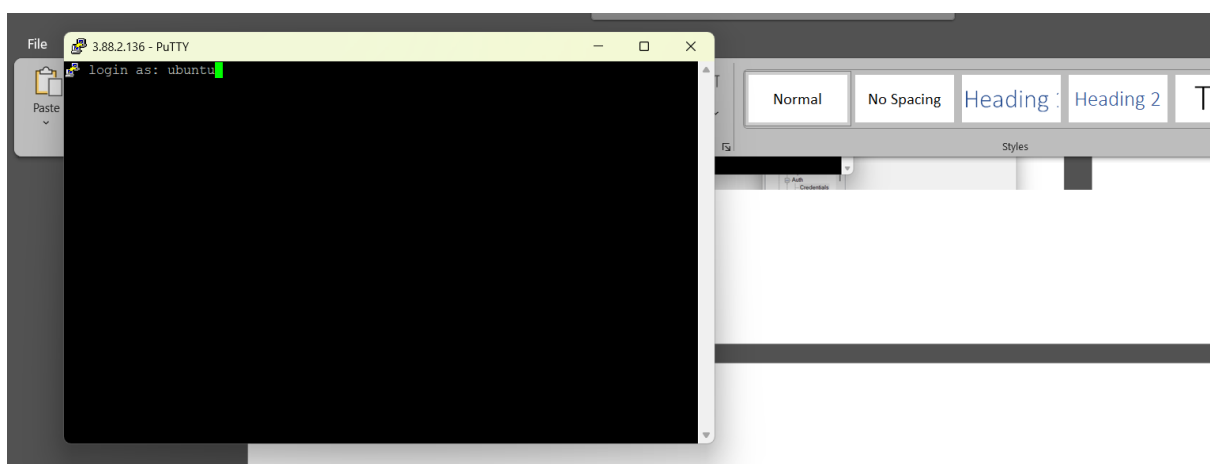
- After Clicking on credentials.
- Choose the private key that we have downloaded.
- And Click Open.



- Terminal will open.



- Now login as ubuntu.



- Now we will create another instance and attached the same EFS file and will check the either we have access to this file or not.
- I have created a new instance 'efs-test'

Instances (3) [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[All states](#)

☐	Name ↗	Instance ID	Instance state ↗	Instance type ↗	Status check	Alarm status	Availability Zone ↗	Public IP
☐	efs-test ↗	i-090363e58e1c82e1f	Running ↗ ↗	t2.micro	Initializing	View alarms +	us-east-1c	ec2-107-
☐	mylinuxvm	i-0b26d636b00c28adb	Running ↗ ↗	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-3-88-
☐	my-vm-serv	i-01a23c7c0adb5b78	Stopped ↗ ↗	t3.micro	-	View alarms +	us-east-1c	-

- Now let's login in the same instance.
- Now I have connected to two different instances.
- Note: Run the commands on both the instances one by one.
 - Sudo su
 - Cd /
 - mkdir efs
- Now go back the EFS console.
- Select the EFS file and click on view details.

File systems (1) [↻](#) [View details](#) [Delete](#) [Create file system](#)

	Name ↗	File system ID ↗	Encrypted ↗	Total size ↗	Size in Standard ↗	Size in IA ↗	Size in Archive ↗	Provisioned Throughput (MiB/s) ↗
☒	efs-demo ↗	fs-04560467dea3abe70 ↗	Encrypted	6.00 KiB	6.00 KiB	0 Bytes	0 Bytes	-

- Click on attach.

Amazon EFS > File systems > fs-04560467dea3abe70

efs-demo (fs-04560467dea3abe70) [Delete](#) [Attach](#)

General [Edit](#)

- Copy the NFS client command.

Attach [×](#)

Mount your Amazon EFS file system on a Linux instance. [Learn more](#)

☒ Mount via DNS ☐ Mount via IP

Using the EFS mount helper:

```
sudo mount -t efs -o tls fs-04560467dea3abe70:/ efs
```

Using the NFS client:

```
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retr=2,noresvport fs-04560467dea3abe70.efs.us-east-1.amazonaws.com:/ efs
```

See our user guide for more information. [Learn more](#)

[Close](#)

- Follow these steps for both the instances.
- Use this command to check the items in the efs directory.
 - `ls -la /efs`

```

root@ip-172-31-10-97:/
root@ip-172-31-10-97:/# ls -la /efs
total 8
drwxr-xr-x 2 root root 6144 Jun 30 14:41 .
drwxr-xr-x 23 root root 4096 Jun 30 14:20 ..
root@ip-172-31-10-97:/#

root@ip-172-31-10-33:/
root@ip-172-31-10-33:/# ls -la /efs
total 8
drwxr-xr-x 2 root root 6144 Jun 30 14:41 .
drwxr-xr-x 23 root root 4096 Jun 30 14:28 ..
root@ip-172-31-10-33:/#

```

- After mounting the EFS, we will create a file in one of the terminals
- Now go to the efs directory
- Create a file using command – `echo "This is a shared EFS file" > /efs/test.txt`

```

root@ip-172-31-10-97:/efs
root@ip-172-31-10-97:/efs# ls -la /efs
total 8
drwxr-xr-x 2 root root 6144 Jun 30 14:41 .
drwxr-xr-x 23 root root 4096 Jun 30 14:20 ..
root@ip-172-31-10-97:/efs# cd /efs
root@ip-172-31-10-97:/efs# echo "This is a shared EFS file" > /efs/test.txt
root@ip-172-31-10-97:/efs#

root@ip-172-31-10-33:/
root@ip-172-31-10-33:/# ls -la /efs
total 8
drwxr-xr-x 2 root root 6144 Jun 30 14:41 .
drwxr-xr-x 23 root root 4096 Jun 30 14:28 ..
root@ip-172-31-10-33:/#

root@ip-172-31-10-97:/efs
root@ip-172-31-10-97:/efs# cat /efs/test.txt
This is a shared EFS file
root@ip-172-31-10-97:/efs#

root@ip-172-31-10-33:/
root@ip-172-31-10-33:/# ls -la /efs
total 8
drwxr-xr-x 2 root root 6144 Jun 30 14:41 .
drwxr-xr-x 23 root root 4096 Jun 30 14:28 ..
root@ip-172-31-10-33:/#

```

- Now we will use the other terminal to check the file in the same directory.
- Here we can see the test.txt file.

```

root@ip-172-31-10-97:/efs
root@ip-172-31-10-97:/efs# cat /efs/test.txt
This is a shared EFS file
root@ip-172-31-10-97:/efs#

root@ip-172-31-10-33:/
root@ip-172-31-10-33:/# sudo su
root@ip-172-31-10-33:/# cd /
root@ip-172-31-10-33:/# ls efs
test.txt
root@ip-172-31-10-33:/#

```

- Here we can also the content.

```

root@ip-172-31-10-97:/efs
root@ip-172-31-10-97:/efs# cat /efs/test.txt
This is a shared EFS file
root@ip-172-31-10-97:/efs#

root@ip-172-31-10-33:/
root@ip-172-31-10-33:/# sudo su
root@ip-172-31-10-33:/# cd /
root@ip-172-31-10-33:/# ls efs
test.txt
root@ip-172-31-10-33:/# cat /efs/test.txt
This is a shared EFS file
root@ip-172-31-10-33:/#

```